Directed Acyclic Graph as Tangle: an IoT Alternative to Blockchains

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***Abstract* — Blockchains are established as the most widely used P2P distributed application for Distributed Ledger Technologies. Nevertheless, their successful existence, especially known due to the cryptocurrency of Bitcoin, gets more and more competitors. One of those competitors is IOTA, which is based on a tangle or a Directed Acyclic Graph architecture. This paper stresses potentials of such architecture, especially in future IoT applications. In particular, few of the most import security issues are analyzed.**

***Key Words* — Directed Acyclic Graph, Distributed Ledger Technology, Tangle, Internet of Things, Transactions, IOTA.**

I. Introduction

DISTRIBUTED Ledger Technology (DLT) are a powerful alternative to the classic database concept: distributed ledger is a database which is independently constructed, held and updated by each participant (node) in a large network. DLT is a relatively new approach for recording, sharing and synchronization of data across multiple data stores. The main difference to the classic or centralized databases is that records contained in a digital ledger (DL) are not communicated by any central authority to various nodes and are shared and synchronized across network by consensus. This is possible due to copies of the ledger, which are made and maintained at each node on the network. The copies of the ledger on all nodes are identical thanks to the process of reaching a consensus every time when the ledger must be updated with new record(s). The synchronization process is fast: any change made to the ledger on one node is copied into the ledgers on other nodes within several seconds or minutes.

All the data in a DL is accurate and immutable due to the use of cryptography, as cryptographic keys and digital signatures are used to secure the access to the ledger's content. Distributed ledgers are very robust to cyber-attacks, because an attacker would need to attack simultaneously the majority of all the distributed copies of a ledger, which is a very hard and complicated task. In case that a single or several copies are corrupted, the integrity of the whole system is not compromised.

There are three basic DLT architectures nowadays: chain or list as blockchain, Directed Acyclic Graph (DAG) as tangle and as tree. Typical representative of the chain or list is a blockchain, mostly known as a base for a cryptocurrency bitcoin invented in 2008 by Satoshi Nakamoto [1].

DAG is a finite directed graph with no directed cycles consisting of finitely many vertices and edges, with each edge directed from one vertex to another, but without back loops. Tangle is a basis for a cryptocurrency IOTA (Internet of Things Association), seen as one of the main future cryptocurrencies in the field of M2M communications. IOTA was founded in 2015 by David Sonstebo, Sergey Ivancheglo, Dominik Schiener and Serguei Popov [2] and works formally as IOTA Foundation since 2017.

Directed Acyclic Graph (DAG) as a tree is the youngest DLT architecture: it is used as the basis for a hashgraph, i.e. a data structure which documents, who and in which order gossiped with whom. Hashgraph was invented by a US professor Leemon Bairds in 2016 [3].

The aim of this paper is to pay attention to a tangle, especially its significant benefits compared to the well-established blockchains.

The paper is organized as follows: Chapter II explains basis architecture mechanisms of tangle,. Chapter III gives an overview of IOTA's properties and concentrates on its benefits, especially compared to blockchains. Chapter IV considers security issues characteristic for DLTs and their cryptocurrencies: negative account balance and double spending problem. Chapter V concludes the paper with the newest directions in the development of the IOTA network.

II. TANGLE

Tangle is a DAG, where each point represents individual transaction, while arrows represent approvals (Fig.1). Each transaction is pointing to the two previous transactions, so called "parents", which were approved by that transaction, so called "child".

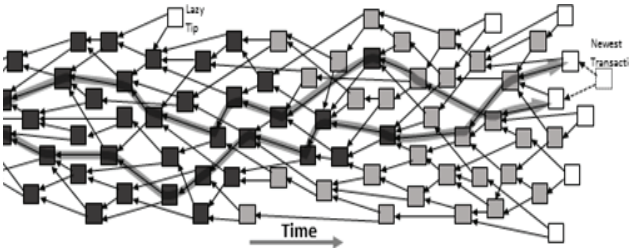

Fig. 1. Tangle: Graph Representation.

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The approvals are marked as arrows on the graph, which are in opposite direction from the direction of the timeline.

A transaction can be approved many times:

- Directly (by the transactions which directly point to it), or
- Indirectly (by all further transactions that may indirectly point to it following the arrows on the graph).

The first transaction in the tangle is the so called "genesis": all IOTA cryptocurrencies were created in the genesis and there is no mechanism to ever create the new ones. Further, all transactions in the tangle approve the genesis directly or indirectly.

The transaction which is approved by large number of newer transactions becomes a part of the consensus and it is practically impossible to change it. This property is known as immutability of transactions and is an inherent DLT property. The consistency of a tangle is accomplished through a small Power of Work (PoW) computation which each transaction has to perform: this makes it too difficult for an attacker to fork or spam the tangle after consensus has been achieved.

The consensus about a valid transaction is reached using the Random Walk Monte Carlo (RWMC) algorithm [4], which is related to the way a new transaction selects two parent transactions to approve. RWMC guides the transaction towards the tips, i.e. to the transactions which are not approved yet and which have as young parents as possible. Nevertheless, there is a small probability to select "lazy tips": the tips chosen to verify some older transactions, which were already verified in the past by some earlier transactions. The selection of a tip from the set of tips is performed by a random walk through graph from the genesis towards tangle's end along the timeline. The walk finishes when a particular tip is selected. The walk is performed twice, so that two tips are chosen for approval and to become the "parents" of the new, "child" transaction.

Cumulative weight of a transaction is the total number of all transactions which came after, and which have approved that transaction directly or indirectly (and increased by 1). The random walking process is guided towards the heaviest branch: in each branching point on the path, the next chosen child transaction is the one which has the largest cumulative weight, but with certain amount of randomness. As the new transactions are coming, the tangle is forced to grow fastest in the directions which follow the paths with the heaviest transactions. In the same time, the transactions appended to the lightest branches are orphaned: most likely they never become a part of the consensus.

III. Benefits of Tangle over Blockchain: M2M Communications

IOTA project aims providing a more suitable cryptocurrency for payments or transactions between billions of future IoT devices, as well as to obtain a reliable platform for many other applications which might include Machine-to-Machine (M2M) communications between those devices. This means, that the implementation of IOTA protocol has to be cheap, fast and scalable, which were the main reasons to develop the tangle instead of using already mature blockchain technology. Increasing number of IoT devices and a rapid development of IoT technology significantly enhanced the demand for micropayments between IoT devices. Bitcoin is, however, not seen as a good choice for two reasons:

- High fees for micropayments, and
- Separation of roles among participants.

Namely, in the Bitcoin society there are miners who verify transactions and "normal users" who issue transactions, whereby miners also may be issuers. With this ambiguity the probability of possible conflicts stays high, and the conflict resolution wastes resources of participants, which cannot be considered as a basis for microtransactions.

Although IOTA network is also a P2P network of nodes like bitcoin network, IOTA nodes are entities which both validate and issue transactions, since before issuing its own transaction a node must validate two earlier transactions issued by other nodes. Furthermore, each node is supported to remain active, i.e. to propagate new transactions from other nodes through the network, although a node perhaps has no more new transactions to issue.

Another quality of IOTA network is its asynchronicity: transaction may be issued at any moment by some node and, after propagation through the network, the transaction reaches different nodes at different moments. This means further that any other node at the moment of issuing its own transaction does not observe the actual state of a tangle, but the one from several moments ago. As a consequence, a node by running tip selection algorithm cannot count on "hidden tips" which would be visible shortly after that moment. Each node refreshes its own version of a tangle. Nevertheless, in general, the versions on different nodes are not necessarily exactly the same: different nodes may see different sets of transactions at any moment. Still, those differences concern at most hidden tips, the newest tips and recently verified transactions, while older transactions which are part of the consensus are common (i.e. the same) for a great majority of nodes.

Nodes are not obligated to maintain always the freshest version of the ledger. They are also not obligated to continuously forward received updates to their neighbours or peers in the network. This means, that it is possible that a node working offline, for example, chooses to verify two parent transactions from an older state of a tangle and, after verification, broadcasts its new transaction through the network. In this way, the node does not approve any new transaction.

Such a lazy behaviour is punished by the protocol in two ways:

- The issued transaction will be considered as a lazy tip from the remaining part of the network, because other active nodes have fresher versions of a tangle, and
- The lazy tips will be most probably orphaned: such transactions will never become a part of the consensus.

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In such a way, nodes are motivated to maintain their ledger versions as much updated as possible. They are also motivated to share new transactions from other nodes since they increase the probability of verifying of their recently issued transactions in that way, and, when verified, also increase the cumulative weights of them, which enlarge the chances to eventually become part of the consensus.

Using the described mechanisms which do not include any transaction fees, tangle offer cheap transactions, unlike Ethereum and other blockchains, which is one of the wanted features for IoT applications. Additionally, tangle is fast and scalable – properties which are also seen as a good IoT support. Unlike blockchain, tangle does not assume transaction blocks, which increases the number of transactions to thousands per second. Finally, tangle is scalable by design since the more transactions are created, the more transactions are validated and confirmed.

IV. SECURITY SUPPORT

Issuing transactions in a tangle also contribute to the tangle's security through resolving possible conflicts: during the verification of two parent transactions, a node checks if the verified transactions are conflicting. Two (or more) transactions can be conflicting in different ways:

- Negative account balance: an entity (a person or an IoT device), purchasing some goods or service, sends IOTA tokens to the other entity (seller) in a transaction, although its account balance calculated from all previous transactions in the tangle (beginning from the genesis event) is negative; this negative balance can be the consequence of the conflict with some earlier transaction(s).
- Double spending problem: a dishonest entity might spend its money twice by issuing two different transactions to different sellers in a short period of time, which results in two conflicting transactions.

It is quite possibly that the tangle contains pairs of conflicting transactions for a while, since there is no mechanism to prevent the appearance of a conflict. Nevertheless, there is a mechanism to combat such problems. Namely, when a node selects two tips (in order to append its child transaction to them), it first has to validate them: it checks the tip's signature and makes sure that the tip is not in conflict with any of the transactions in its validation path, i.e. with the transactions which are directly and indirectly validated by this tip. In case that the node finds out that the selected tip is in conflict, the node abandons that tip and chooses another one. This is the direct way of conflicts resolving.

Similarly, if the node finds out that there is a conflict between any two transactions in the validation path of the selected tip, the node abandons that tip and chooses another one. In this case the conflict is indirectly resolved. In case that a node would issue a new transaction, which would indirectly approve conflicting transactions, there would be a risk for the node that its new transaction would not be approved by other nodes.

V. RECENT AND FUTURE DEVELOPMENT

One of the biggest disadvantages of IOTA, which was very often stressed, was that the tangle was not decentralized, as claimed to be. Indeed, the existence of a network "coordinator", who controlled the growth of the network and solving of forking problems, was a significant drawback of the network and a limiting factor of the network scalability.

Nevertheless, the recent news about IOTA's decision to remove the coordinator role from the platform is a big step forward to decentralization. With such a new development, IOTA's tangle is also a big step forward to becoming a strong DLT for IoT and M2M purposes.

VI. FUTURE WORK

The paper presents a concept of distributed acyclic graphs and IOTA as a promising alternative to the well-established blockchain technology. An overview of a tangle technology, as well as its security aspects are presented.

The future work on this topic will include implementation of the technology for a specific use case as a proof of concept, as well as evaluation and testing of the proposed concept.

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